



SURFACE VEHICLE STANDARD

J1757™-2

NOV2018

Issued 2018-11

Standard - Optical System HUD for Automotive

RATIONALE

This SAE Standard defines a consistent terminology and metrology for vehicular Head Up Display (HUD). It allows measurement of automotive HUD attributes in accordance to SAE J1757-1 and ISO 15008.

NOTE: All definitions and units are for the metric system (SI) unless noted otherwise.

1. SCOPE

This SAE Standard provides measurement methods to determine HUD optical performance in typical automotive ambient lighting conditions. It covers indoor measurements with simulated outdoor lighting for the measurement of HUD virtual images. HUD types addressed by this standard includes w-HUD (windshield HUD) and c-HUD (combiner HUD) with references to Augmented Reality (AR) HUD as needed. It is not the scope of this document to set threshold values for automotive compliance; however, some recommended values are presented for reference.

2. REFERENCES

2.1 Applicable Documents

The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the latest issue of SAE publications shall apply.

2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

SAE J941 Motor Vehicle Drivers' Eye Locations

SAE J1757-1 Standard Metrology for Vehicular Displays

2.1.2 ISO Publications

Copies of these documents are available online at <http://webstore.ansi.org/>.

ISO 4513 Road Vehicles - Visibility - Method for Establishment of Eye-Ellipses Location

ISO 15008 Road Vehicles - Display Legibility Standard

ISO 16505 Road Vehicles - Ergonomic and Performance Aspects of Camera-Monitor Systems - Requirements and Test Procedures

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SAE WEB ADDRESS:

2.2 Related Publications

The following publications are provided for information purposes only and are not a required part of this SAE Technical Report.

2.2.1 CIE Publications

Available from CIE Central Bureau, Babenbergerstrasse 9/9A, 1010 Vienna, Austria, Tel: +43 1 714 31 87, www.cie.co.net.

- CIE 85 Technical Report - Solar Spectral Irradiance
- CIE 15.2 Recommendations on Uniform Color Spaces, Color Differences Equations, Psychometric Color Terms
- CIE 17.4 Recommendations - CIE International Lighting Vocabulary
- CIE S003/E CIE Standard - Spatial Distribution of Daylight - CIE Standard Overcast Sky and Clear Sky

2.2.2 ICDM Publications

Available from SID organization at SID - Society of Information Display, 1475 S Bascom Ave, Ste 114, Campbell, CA 95008; Tel: (408) 879-3901, www.sid.org; or directly on the ICDM webpage, www.icdm-sid.org.

- IDMS1 Information Display Measurements Standard

2.2.3 U.S. Government Publications

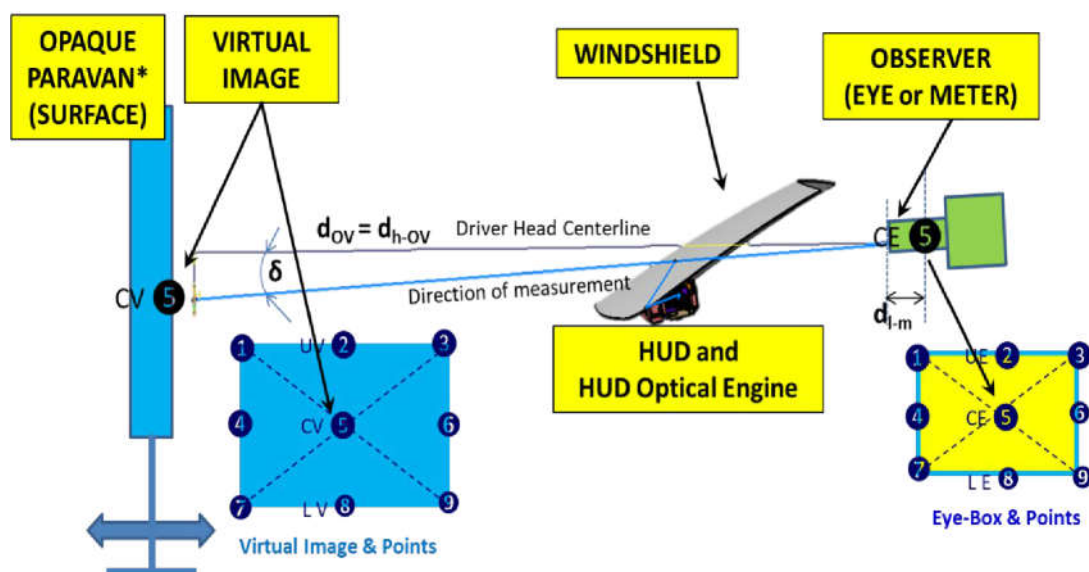
Copies of these documents are available online at <http://quicksearch.dla.mil>.

- MIL-87213 Military Handbook - Electronically/Optically Generated Airborne Displays

3. DEFINITIONS

3.1 THE DRIVER HEAD CENTERLINE

A longitudinal reference line that passes through the centroid of Eyellipse along the longitudinal axis (x Axis) of the vehicle. See Figure 2.



(a)

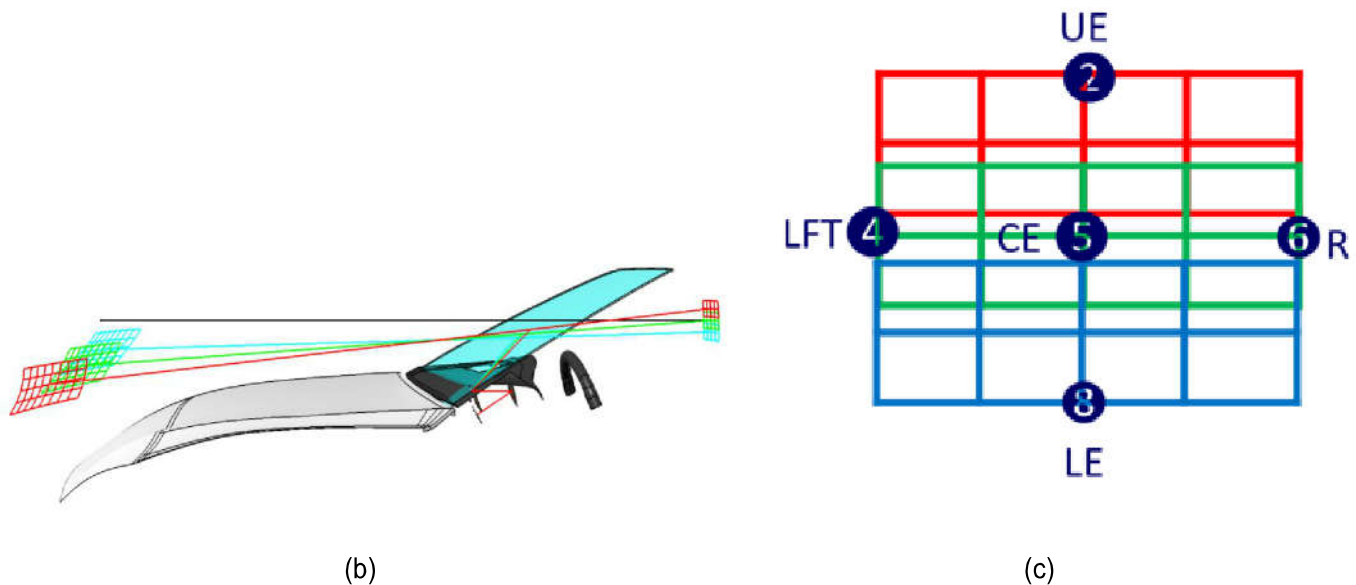


Figure 1 - System HUD and setup for measurements

(a) Overview of the setup

(b) Illustration of driver's perception of the virtual image, as seen from three eye positions

(c) Illustration of the measuring points and their relationship to traditional HUD eye-boxes

3.2 EYELLIPSE

The statistical distribution of driver's eye locations relative the SAE SgRP (H-pt) as defined by SAE J941.

3.3 EYEBOX

A vertical plane projected through the centroid of an Eyellipse (see Appendix A for details). The Eyebbox width and height limits are the Y (cross-car) and Z (Vertical) boundaries of the Eyellipse and used to define where the HUD virtual image can be seen by the population of driver's eyes. (See Figure 2).

3.4 PARAVAN

A vertical screen (physical object) with orthogonal horizontal (Y) and vertical (Z) rulings (graduated perpendicular lines) used to evaluate HUD virtual image qualities like luminance, size, color, virtual image distance from the Eyebbox, X and Z adjustment ranges, and HUD Field of View (FOV).

3.5 MEASURING POINTS FOR AN OBSERVER OR METER

Defined by the Eyebbox; see Figures 2 a1 and a2 (Observer / Instrument Meter Locations) for details. The CE5 (Center), UE2 (Upper), LE8 (Lower), LFT 4 (Left), and R6 (Right) are the Eyebbox measuring points and describe the front side of the sensor surface for positioning of optical instrumentation. The standardized measurement points for the virtual images are located on the paravan, as shown in Figure 1.

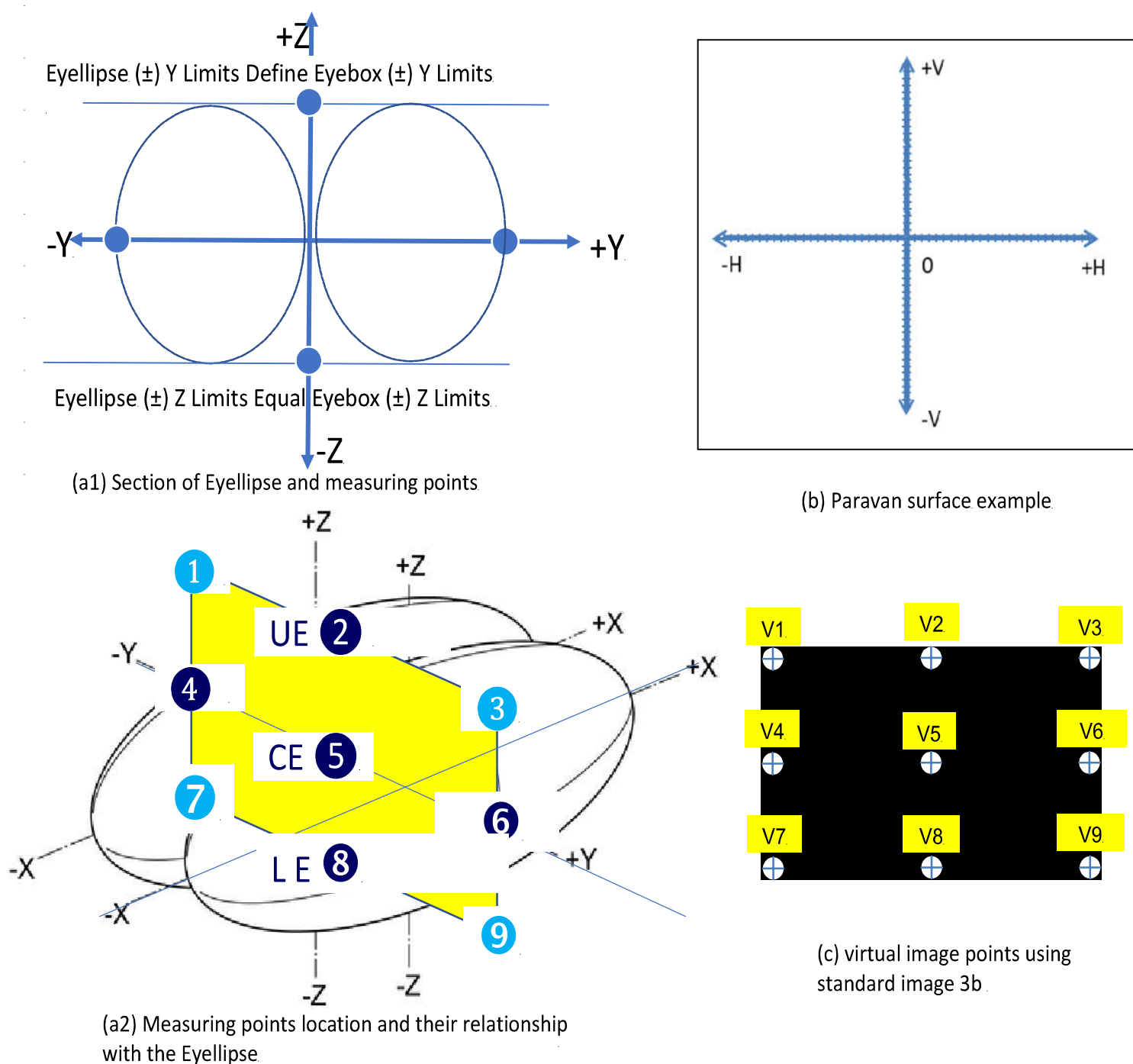
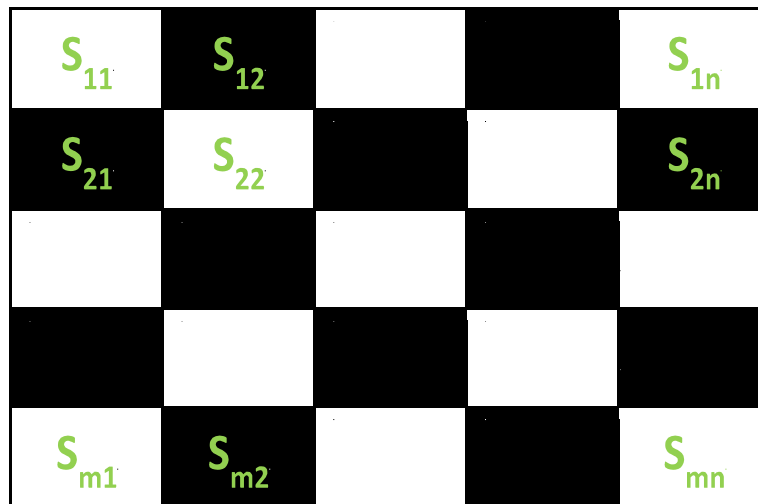


Figure 2 - Illustration of measuring points location

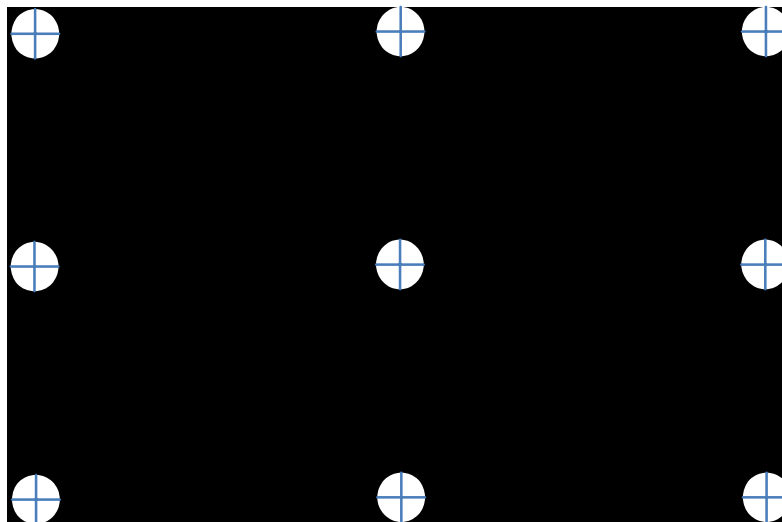
- (a) a1 and a2 illustrations are related to the Eyellipse (includes cross-section shown in a1)
- (b) Coordinate system on the paravan surface example
- (c) virtual image points using standard image 3b

3.6 SYSTEM HUD ORIGINAL IMAGE SOURCE

The HUD's optical engines active area and generates the virtual image (see Figure 1). This standard image is either a set of even squares (as shown in Figure 3a for $n=m=5$), or a set of one-surface dots with a minimum of three dots per line (as shown in Figure 3b for $n=m=1$). Full white and full black images should be used for optical performance measurements, as defined by CIE.



**Figure 3a - Standard Image for Luminance measurements
option (a) - shown for $n=5$; $m=5$.**



**Figure 3b - Standard image for measurements
option (b) - the circles are tangent to the edges of the image**

Figure 3 - Standard images options

3.7 FIELD OF VIEW (FOV)

For a HUD's virtual images active area, FOV is defined as its size in horizontal (X) and vertical (Z) dimensions, and measured in degrees (from the CE5 point). This solid cone, with its apex at the centroid of the Eyellipse, is at point CE5. Therefore, the FOV is the combined angular measurements of the vertical (Z) and horizontal (Y) angles. Points V2, V8, V4, and V6 on the target image determines the measured FOV. See Figure 1 for details.

3.8 OBSERVER-EYE LOCATIONS

Defined by the Eyellipse, Eyebox, and the measurement points, as shown in Figure 2.

3.9 MEASUREMENTS

Defined by the points located on the observer/instrument meter surface and points on the virtual image. See Figure 1 for details.

3.10 VIEWING ANGLE (δ)

A side-view angle using the centroid of the Eyellipse/Eyebox as the horizontal datum to measure viewing angles up or down from this reference plane.

3.11 DISTANCE FROM OBSERVER TO VIRTUAL IMAGE ($d_{ov} = d_{h-ov}$)

Defined from center of the Eyebox to the virtual image on the paravan surface. Since the windshield is between the observer and the virtual image, the virtual image distance to windshield (d_{wv}) and distance between windshield to observer (d_{ow}) is considered in these calculations. See Figure 1 for details.

$$d_{ov} = d_{ow} + d_{wv} \quad (\text{Eq. 1})$$

NOTE: The optical path for light when the observer-to-virtual image measuring distance is different than the quantitative distance is not considered in this standard.

3.12 LUMINANCE OF THE VIRTUAL IMAGE

Defined as the luminous flux (Φ) emitted by the virtual image from a surface per unit solid angle per unit area in an Eyebox direction. The SI unit is candela per square meter (cd/m^2), the English unit is footlambert (fL) ($1 \text{ fL} = 3.4263 \text{ cd/m}^2$). Refer to SAE J1757-1.

3.12.1 $L_{[\text{dark},w]}$ = Luminance for the virtual image white icon (information) is measured in the dark ambient light (ambient light is OFF).

3.12.2 $L_{[\text{dark},k]}$ = Luminance for the virtual image black icon (background) is measured in the dark ambient light (ambient light is OFF).

3.13 CONTRAST (C)

Defined as:

$$C = (L_{(\text{information})} - L_{(\text{background})}) / L_{(\text{background})} \quad (\text{Eq. 2})$$

For contrast measurements, refer to SAE J1757-1 Table 1A.

3.14 CONTRAST RATIO (CR)

The ratio between the information luminance and background luminance:

$$C = L_{(\text{information})} / L_{(\text{background})} \quad (\text{Eq. 3})$$

Contrast (C) and contrast ratio (CR) are related by the formula:

$$C = CR - 1 \quad (\text{Eq. 4})$$

Contrast ratio in high ambient light illumination is defined as CRd for diffuse (skylight) illumination and CRh for direct sunlight illumination. Refer to SAE J1757-1 Table 1 in 4.1 for more information.

For this document, the information is considered white color ($L_{\text{(information)}} = L_{\text{[dark,w]}}$) and the background is dark color ($L_{\text{(background)}} = L_{\text{[dark,k]}}$). See contrast (3.13).

NOTE: The contrast ratio value is critical to determine display legibility in all ambient light conditions (daylight with and without direct sunlight and night). Contrast (and CR) at any ambient is influenced by road/ambient illumination and reflections toward the display, and influences both background and measured information luminance.

3.15 HUD IMAGE RESOLUTION

Depends on original optical engine and magnifications. It is defined in H and V direction as the number of optical engine pixels (or equivalent pixels)/degree.

$$\text{HUD } H_{\text{Res}} = P_{x-H} / \text{HUD } H_{\text{angle}} \quad (\text{Eq. 5a})$$

$$\text{HUD } V_{\text{Res}} = P_{x-V} / \text{HUD } V_{\text{angle}} \quad (\text{Eq. 5b})$$

$$\text{HUD }_{\text{Res}} = \min\{\text{HUD } H_{\text{Res}}, \text{HUD } V_{\text{Res}}\} \quad (\text{Eq. 6})$$

Where P_{x-H} and P_{x-V} are respectively the number of horizontal and vertical pixels the optical engine or source that contribute to the virtual image.

See also active area-source and number of pixels.

3.16 ACTIVE AREA

For a HUD's optical engine, there is the area covered by pixels or equivalent pixels that creates the virtual image. Refer to SAE J1757-1.

3.17 DISPLAY ADDRESSABILITY

Expressed in $P_h \times P_v$ (number of horizontal pixels x number of vertical pixels), such as VGA (640 x 480) or SVGA (800 x 600).

3.18 ASPECT RATIO (W:H)

The active area width-to-height ratio (such as 4:3 for NTSC displays, or 16:9 for wide view displays).

3.19 CANDELA (cd)

The unit of luminous intensity (I) and is the luminous flux (Φ) in lumens (lm) emitted per unit solid angle (lm/sr).

3.20 SYSTEM HUD VOLUME

The total volume occupied within the instrument panel/cockpit area.

4. MEASUREMENTS

4.1 System HUD Virtual Image Measurements

A measuring device for luminance, uniformity, and color must be NIST (or National Standard Lab) traceable, with a 5% expanded uncertainty, with a coverage factor of two for luminance and ± 0.008 for color (UV); refer to SAE J1757-1 for details. Distance between observer and paravan must be measured with an uncertainty similar or better than optical meter requirements of 5%.

The following set up steps are necessary for any measurements.

Step 1: HUD Location

- a. Set up the HUD system in a lab using vehicle location coordinates for all optical elements, like optical HUD engine, mirror system, windshield, etc. Location of all optical elements must be accurate and repeatable, and documented in reported measurements.

Step 2: Setup of Measuring Box (Eyebox) and Meter Position

- a. Create a fixture capable of locating measuring devices at the specified Eyebox measuring points, as defined in Figure 2.
- b. Locate the measuring devices at the specified measuring points within the Eyebox per Figures 1 and 2. These observer measuring points are located at the front side of the sensor for meters or cameras.

Step 3: Set Up Metrology System for Virtual Image Performance Measurements

- a. Create a movable fixture, such as an opaque monocolored paravan, with a square area or ruler printed in centimeters (see Figure 2b).
- b. Use the HUD to display a standard image, such as shown in Figure 3b.
- c. Position the measuring device in the point CE5 (Eyellipse and Eyebox centroid) located on the front surface of the meter (or camera), and focus on the virtual image at point CV5. See Figure 2.
- d. Move the opaque monocolored paravan so both the virtual image and grating (vertical and horizontal rulers) are in focus. "In focus" is determined by a device with focusing capability, such as a luminance meter or camera. See Appendix B.
- e. Adjust the HUD system and meter (or camera) location so the center of the ideal image (or target value, shown in yellow color in this document) corresponds to the center of the measured HUD system sample image (or measured value, shown in red color in this document).

4.1.1 Measure the Distance to Virtual Image (d_{ov})

Perform the Steps 1 to 3 as described above, and then perform Step 4.

Step 4: Measurement of the HUD Virtual Image Distance

- a. The distance between the point CE5 and the paravan is the HUD distance of the virtual image at V5 (d_{h-ov}). See Figure 1 for details.

NOTE: If a measuring device capable of converting focal distance to real distance is used, and the measurement is within acceptable values the real distance is the value shown by the meter. The focal distance depends on lens, aperture, and binocular human vision for estimation. See Appendix B, which discusses the dependence of the depth of field on the aperture size when measuring virtual image distances.

4.1.2 FOV Measurements

Perform 4.1.1 Steps 1-3 using HUD standard image as shown in Figure 3b.

Step 1: Measure Horizontal FOV

- a. Display the standard image in the HUD, as shown in Figure 3b.
- b. Measure the distances between V1 and V3, V4 and V6, and V7 and V9.
- c. Average the three distances to attain d_{FOV-H} .
- d. Use the following formula to convert horizontal distance into horizontal FOV:

$$\alpha/2 = \text{Arctan} ((d_{FOV-H}/d_{ov})/2) \quad (\text{Eq. 7a})$$

Step 2: Measure Vertical FOV

- Display the standard image in the HUD, as shown in Figure 3b.
- Measure the distances between V1 and V7, V2 and V8, and V3 and V9
- Average the three distances to attain d_{FOV-V} .
- Use the following formula to convert vertical distance into vertical FOV:

$$\beta/2 = \text{Arctan} ((d_{FOV-V}/d_{OV})/2) \quad (\text{Eq. 7b})$$

α (alpha) is the angular FOV for horizontal (Y) direction and β (beta) is the angular FOV for vertical (Z) direction. In real life, the angle δ (delta) between horizontal direction (determined by d_{h-OV}) and distance direction (determined by d_{OV}) is very small (several degrees), and length is high (>2 m range), so trigonometric approximations can be used. In other cases, the geometrical conversions should be considered. See Figures 1 and 4 for details.

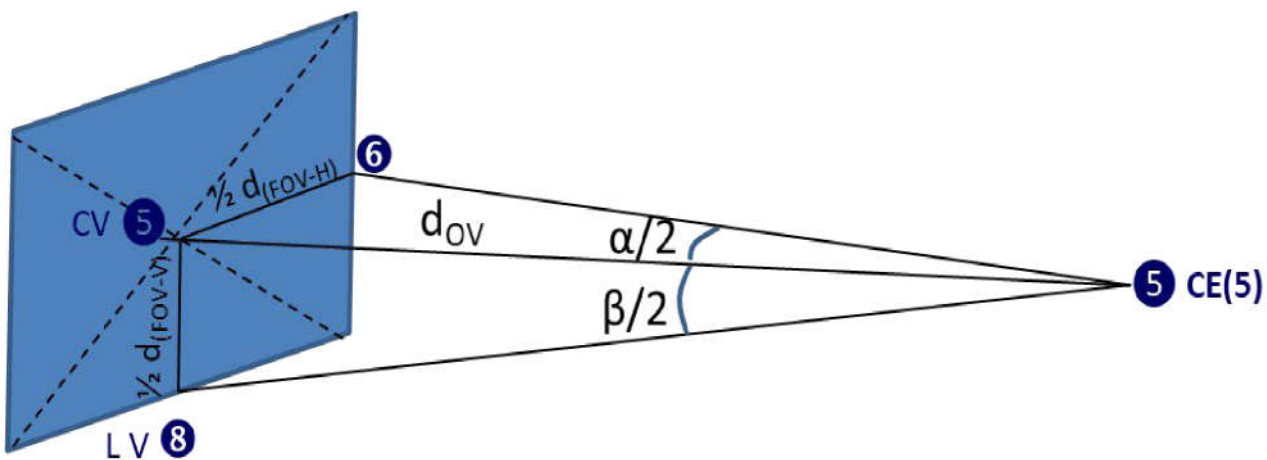


Figure 4 - Measuring examples

NOTE: Conversion from “size” to angular measurements (assuming d_{OV} is perpendicular to paravan due to a small δ angle between the d_{OV} and d_{h-OV} directions). For large δ angles, geometric conversion should be used.

4.1.3 Luminance Measurements

- Set up the sytem HUD as per 4.1.
- Create the dark room environment condition as per SAE J1757-1 Table 1/ISO 15008 (less than 50 lx).
- Display standard images, such as luminance checkerboard (see Figure 3a), with white in the center and then with black in the center. Switch between them to measure L_{white} and L_{black} .
- Position the measuring device in the CE5 location and measure $L_{background}$ and $L_{information}$. Use SAE J1757-1 Table 1A and 4.1 of this standard to determine luminance values and calculate CR as in Eq. 4.

NOTES:

- It is suggested to use a full white and full black pattern. If a spot meter is used, the spot sensor area must meet optical requirements for usage (typically about nine measuring pixels corresponding to one pixel image; refer to IDMS1 reference for more details).
- Eliminate stray light issues for accuracy improvements (refer to IDMS1 reference for details).

3. If a paravan checkerboard is used for virtual image measurements, calculate average luminance for virtual image (AVG L_{vi}) for all white L_{ij} areas (i=1 to n; j=1 to m).
4. Virtual image appearance may be influenced by polarized eyeglasses, so this can be added to the measurements, if desired.

4.1.4 Uniformity Measurements.

- a. Set up the sytem HUD as per 4.1.
- b. Measure luminance uniformity as per SAE J1757-1.
- c. The measurement should be performed in a dark room (refer to SAE J1757-1 Table 1A). Recommended ambient illumination is 10 lx.

Step 1: Display the Full Screen Virtual Image

- a. Black or gray level 0 of the system for the LNUBK.
- b. White or maximum gray level of the system for white luminance non-uniformity LNUW.
- c. Both values should be measured.

Step 2: Perform the Measurement

Use the measurement points defined in Figure 5. The measuring meter center spot locations are shown in Figure 5. The diameter of the spot size range should be between 0.2V and 0.2H. The preferable value is the minimum of 0.2V and 0.2H. The minimum spot size should cover at least 25 pixels on the imaging device.

Step 3: Calculate LNU Based on Formula (Eq. 8)

$$LNU = [(L_{\max} - L_{\min}) / L_{\max}] \times 100\%$$

(Eq. 8)

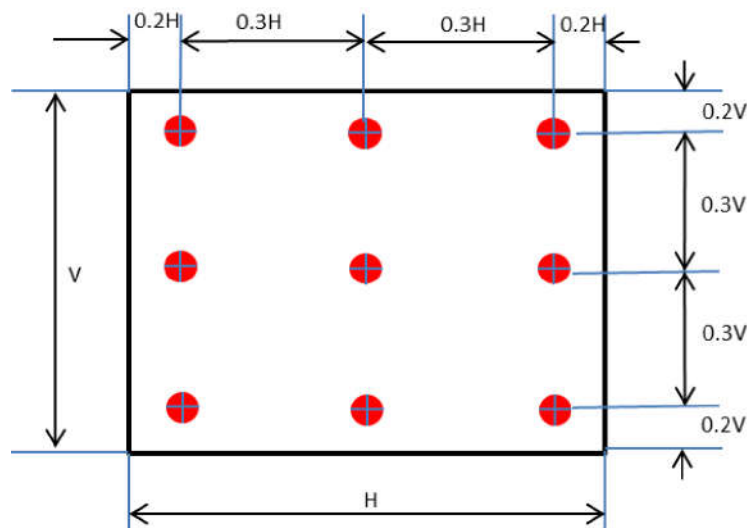


Figure 5 - Points for non-uniformity measurements (not to scale)

NOTE: Intended for all points of measurements

4.1.5 Rotation Measurements

- Display the standard image in the HUD, as shown in Figure 3b.
- Locate the theoretical points 4 and 6, either physically on the paravan, or with digital data in the camera software.
- Measure the location of points V4 and V6 and calculate the vertical d_v and horizontal d_h distances between the points.
- Image rotation should be calculated using the following formula:

$$\varnothing = \text{Arctan} (d_v/d_h) \quad (\text{Eq. 9})$$

where:

$\varnothing$ = angle of virtual image rotation

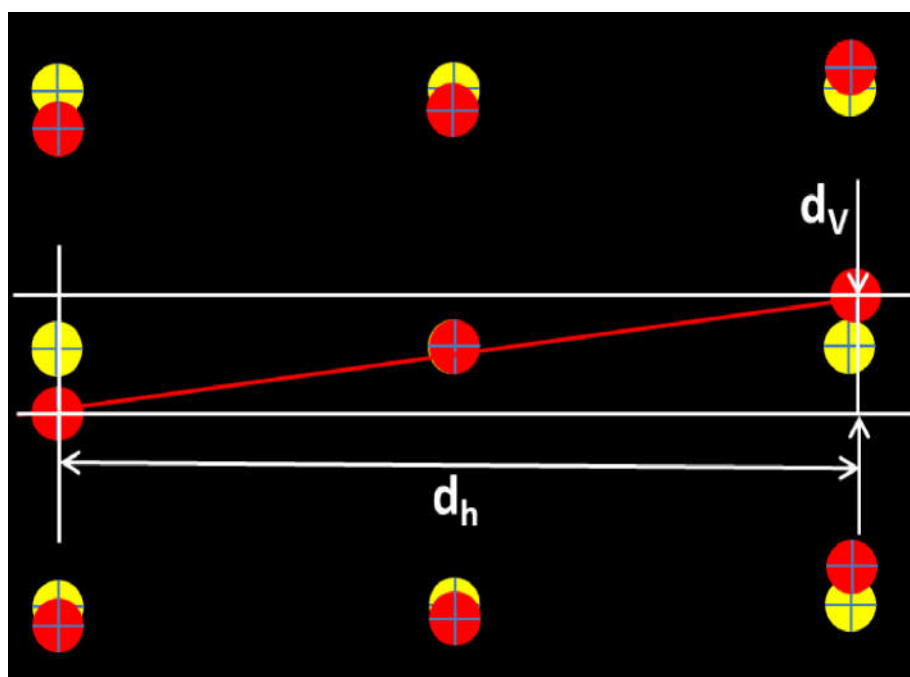


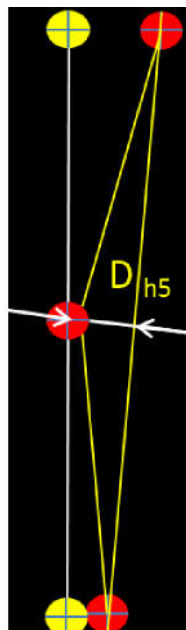
Figure 6A - HUD rotation

NOTE: The virtual ideal image (target - yellow collar) shown, and HUD sample image shown (red color here).

4.1.6 Distortion Measurements

4.1.6.1 Horizontal Distortion Measurements

- Display the standard image in the HUD, as shown in Figure 3b.
- Measure the position of points 2, 5, and 8, either physically on the paravan, or with digital data in the camera software, as shown in Figure 6B.



NOTE: D_{h5} is the horizontal distortion for point 5 and is the absolute value of the distance between point 5 and the line determined by points 2 and 8.

- Repeat measurements for points 1, 4, and 7 and 3, 6, and 9 to determine D_{h4} and D_{h6} , respectively.
- The total horizontal distortion is D_{ht} :

e. The maximum horizontal distortion value is D_{hmt} :

NOTE: Instrument meter is located in point CE5 for this example. The additional instrument meter locations should be in points 2, 4, 6, and 8 to repeat the distortion measurements.

4.1.6.2 Vertical Distortion Measurements

- Display the standard image in the HUD, as shown in Figure 3b.
- Determine the position of points 4, 5, and 6, either physically on the paravan, or with digital data in the camera software.

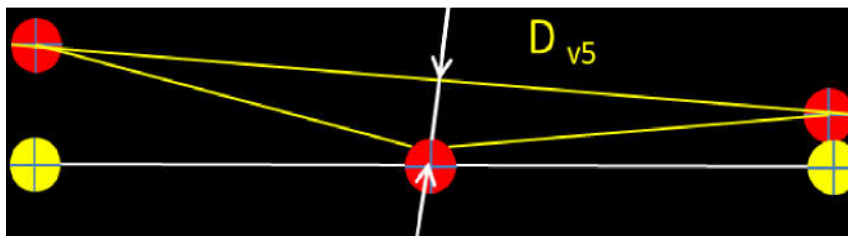


Figure 6

NOTE: D_{v5} is the vertical distortion for point 5 and is the absolute value of the distance between point 5 and the line determined by points 4 and 6.

- c. Repeat measurements for points 1, 2, and 3 and 7, 8, and 9 to determine D_{v2} and D_{v8} , respectively.
- d. The total vertical distortion is D_{vt} :

$$D_{vt} = \text{Avg} \{D_{v2}, D_{v5}, D_{v8}\} \quad (\text{Eq. 10c})$$

- e. The maximum vertical distortion value is D_{vmt} :

$$D_{vmt} = \text{Max} \{D_{v2}, D_{v5}, D_{v8}\} \quad (\text{Eq. 10d})$$

NOTE: Instrument meter is located at point CE5 for this example. Determine the position of points 2, 4, 6, and 8 to repeat the distortion measurements.

4.1.7 Ghost Image Measurements

Perform 4.1.1 Steps 1 through 3 using HUD standard image, as shown in Figure 3b.

- a. Display the standard image in the HUD, as shown in Figure 3b.
- b. Looking from point CE5, find the ghost image for point V5.
- c. Measure the distance from point V5 and the ghost of point V5 ($D_{ghost, CE5}$).

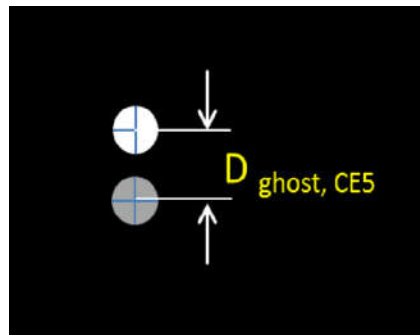


Figure 7 - Ghost measuring example shown for a measuring point

- d. Repeat measurements looking from points UE2 and LE8 and measure distances $D_{ghost, UE2}$ for V2 and $D_{ghost, LE8}$ for V8, respectively.

Overall ghost image is $D_{ghost, t}$:

$$D_{ghost, t} = \text{Avg} \{D_{ghost, CE5}, D_{ghost, UE2}, D_{ghost, LE8}\} \quad (\text{Eq. 11})$$

NOTES:

1. Ghost images will not be as bright as the primary image, so the aperture of the camera may require adjustments in order to see the ghost images.
2. Ghost image distances will vary depending on the viewing locations, so measurements should be performed meters at the upper point UE2, central point CE5, and lower point CE5.

5. NOTES

5.1 Revision Indicator

A change bar (I) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

PREPARED BY VEHICULAR FLAT PANEL DISPLAY AT STANDARDS COMMITTEE

APPENDIX A

A.1 EYELLIPSE DEFINITION AND MEASURING POINTS LOCATION

ISO 4513:2010/SAE J941 establish tangent cutoff Eyellipse percentiles for 2D- and 3D-inclusive Eyellipse percentiles for adult 50/50 male/female drivers in vehicles with and without adjustable seats.

NOTES:

1. Neck pivot, P, points are defined to establish specific left and right eye points for direct and indirect viewing tasks, as described in SAE J1050. These P points are defined only for adjustable seat eye ellipse.
2. ISO 4513:2010 applies to Class A vehicles (passenger cars, multipurpose passenger vehicles, and light trucks), as defined in SAE J1100. It also applies to Class B vehicles (heavy trucks).

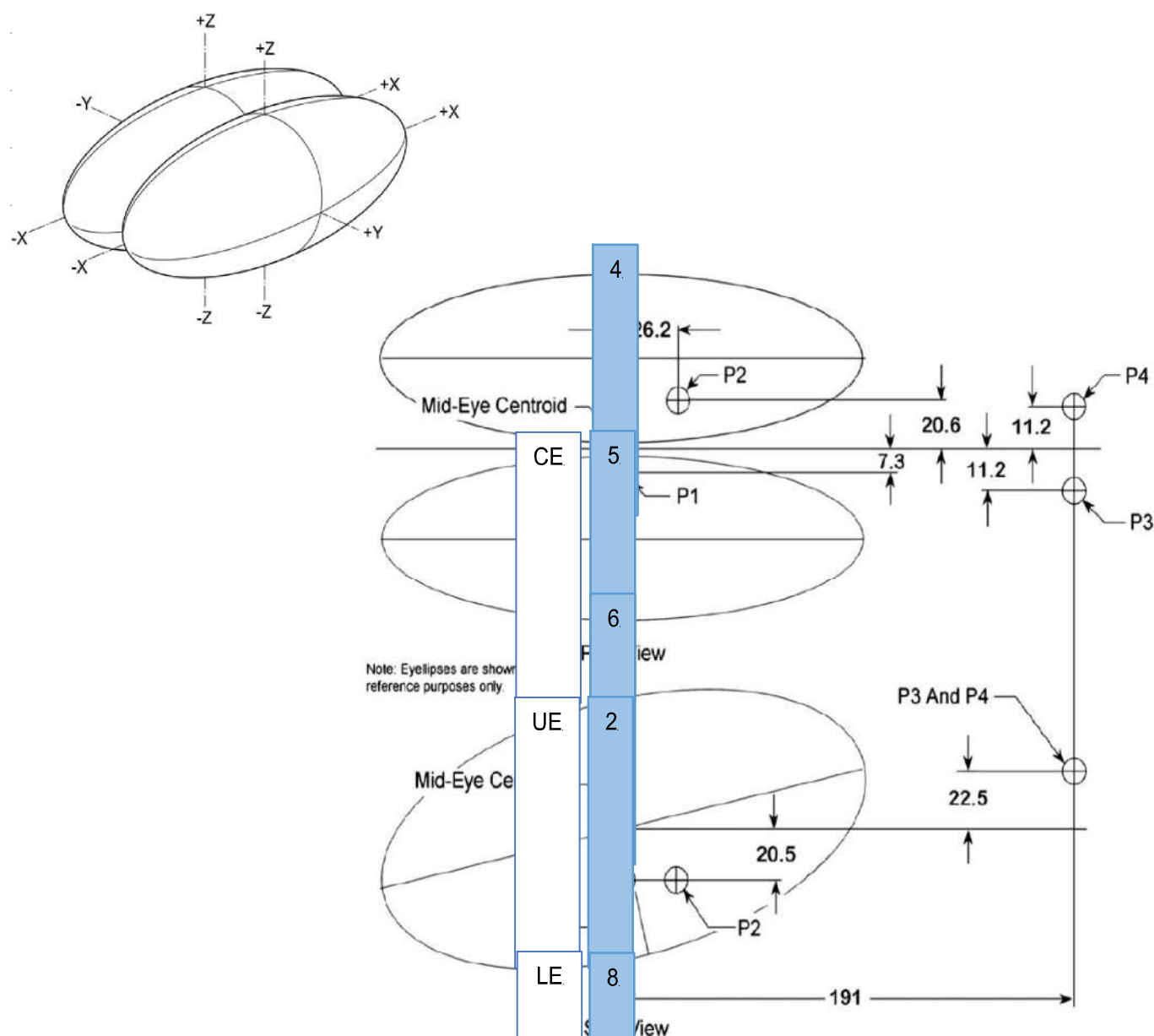


Figure A1 - P-point location relative to 95th percentile eyellipse mid-eye centroid for track travel >133 mm

The following reference figure is from SAE J941, and is used to illustrate that the Eyebbox Z dimension is created by projecting a vertical plane through the centroid of the Eyellipse. A 95th percentile Eyellipse is shown, but this technique is applicable for any size Eyellipse.

NOTE: All dimensions are approximate and intended to communicate relative differences between the Eyebbox height versus the overall 95th Eyellipse side view height.

In this example, the 95th Eyebbox Z dimension is about 3 mm less at the top and bottom of the side view 95th Eyellipse height, because the Eyellipse has a 12-degree side view angle. The Eyebbox is full width of the plan view Eyellipse because it is parallel to centerline of driver. Eyellipse calculations or CAD solutions can be used to define any Eyebbox size, so the purpose of this illustration is to show how to adjust Eyebbox heights (Z) relative to a chosen Eyellipse.

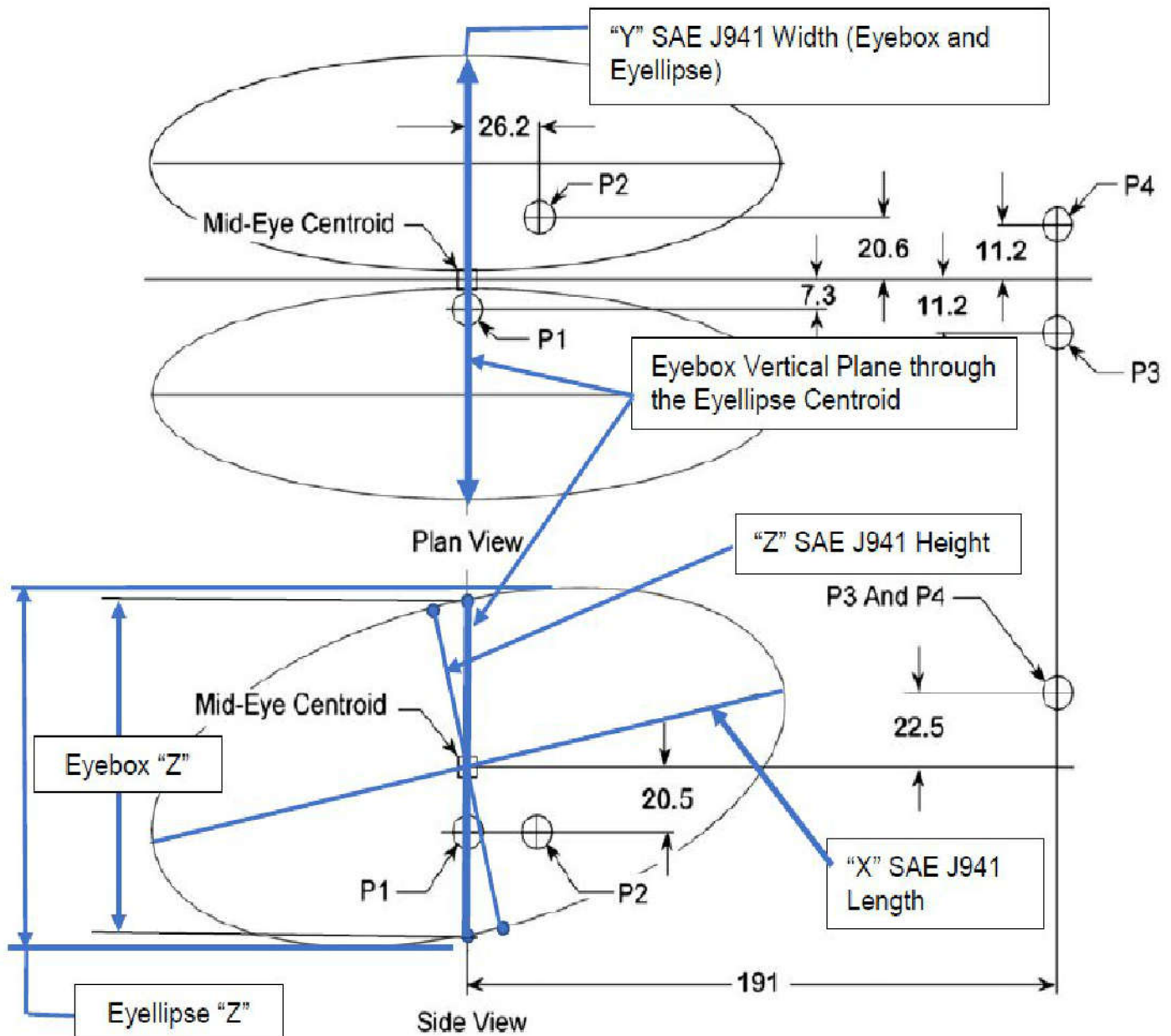


Figure A2 - P-point locations relative to 95th percentile eyellipse mid-eye centroid with seat track travel >133 mm

A.1.1 SAE J941 Clarifications on Using the Eyellipse and Eyebox for HUD Measurement Points

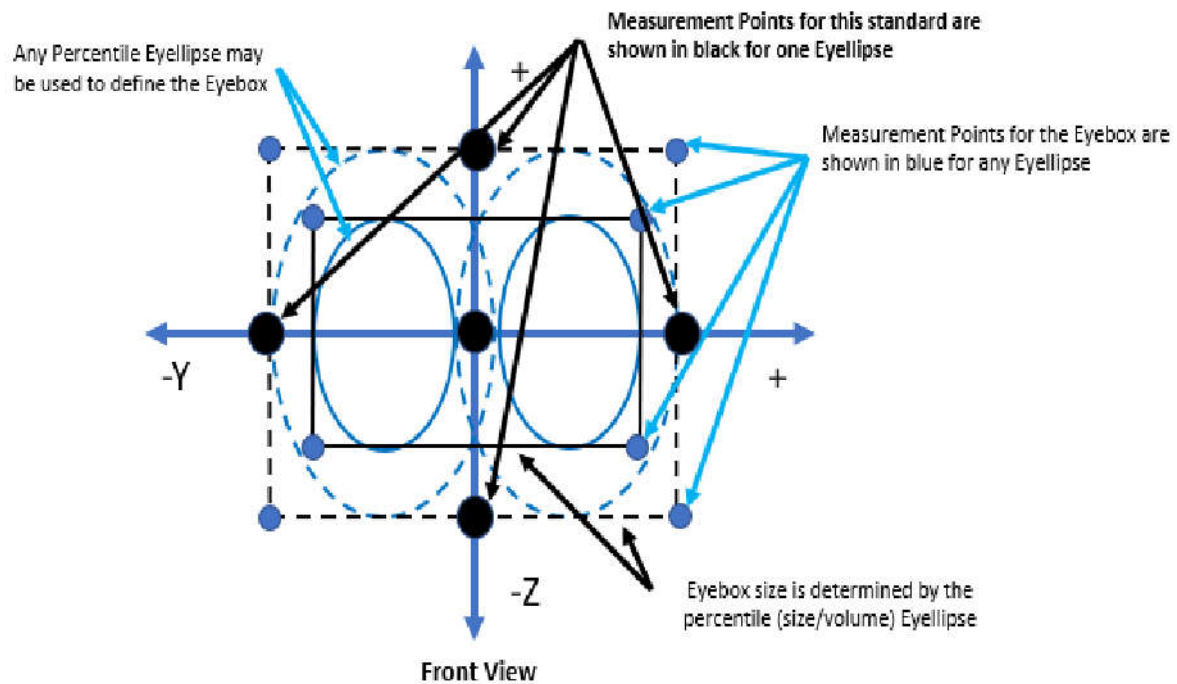


Figure A3 - Clarifications on using the eyellipse and eyebox for HUD measurement points, front view

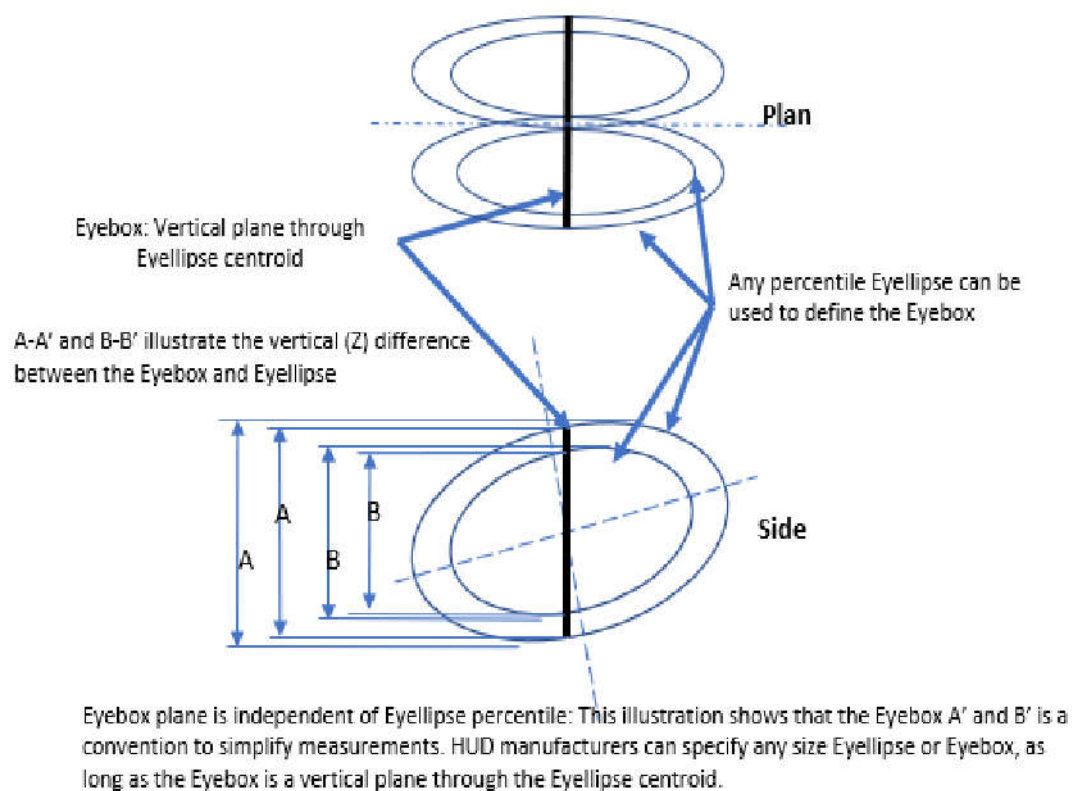


Figure A4 - Clarifications on using the eyellipse and eyebox for HUD measurement points, plan and side view

APPENDIX B

B.1 REFERENCE AND EXAMPLE FOR MEASUREMENTS

NOTE: Company name was mentioned as a source for data.

USEFUL BACKGROUND

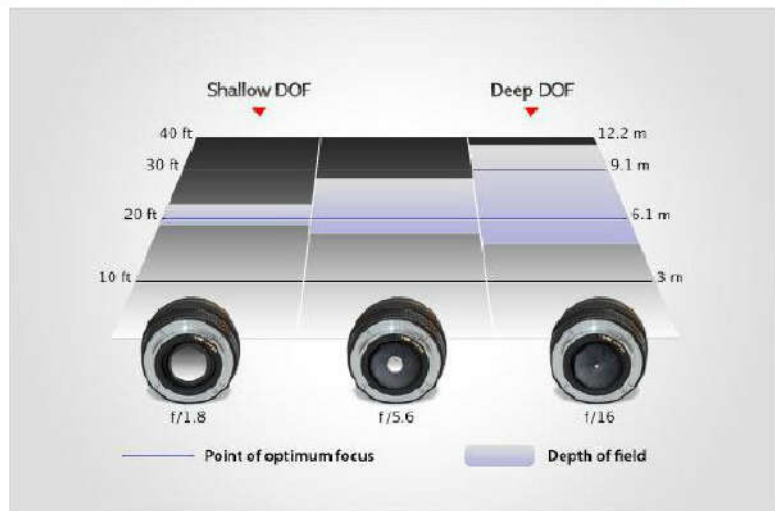
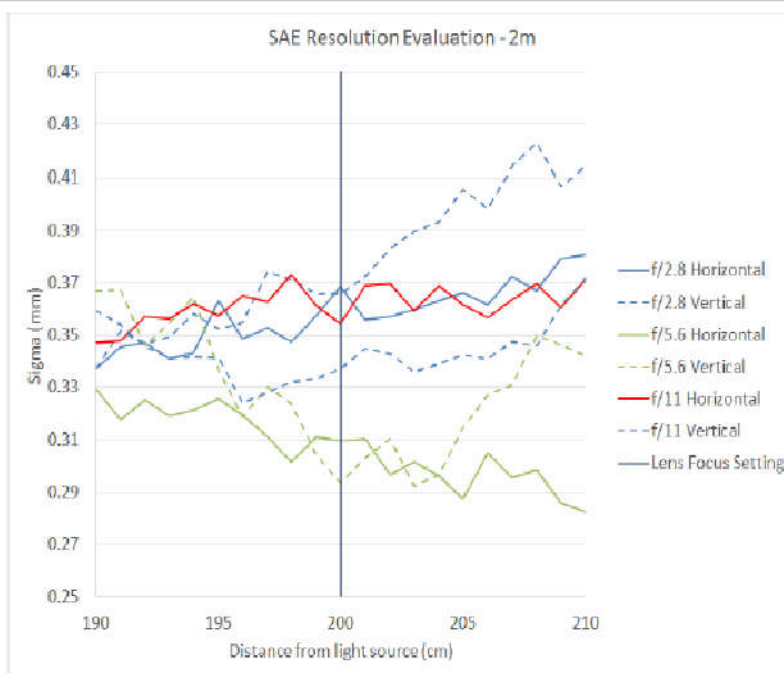


Image courtesy of "Focusing Basics", Exposure Guide
<http://www.exposureguide.com/focusing-basics.htm>

It is well known in photography that the depth of focus (or field) is dependent upon the aperture setting of the lens, as well as the focus distance to the subject.

RESULTS: WD = 200cm



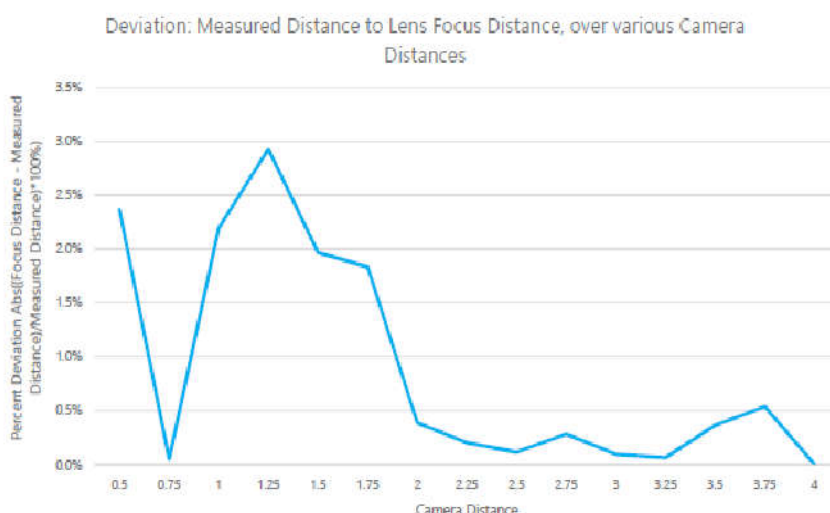
- When the *target* working distance was 2m, the quality of focus of the scene varied very little as a function of measured working distance when the iris was set to $f/2.8$, $f/5.6$ and $f/11$.
- This suggested that the depth of focus of the imaging system was not fully explored at 2m while using a range of ± 10 cm centered on each target distance.

Figure B1

B.2 TEST PROCEDURE - DISTANCE USING DIFFERENT CONTRAST RATIOS (TEST BY RADIANT EXAMPLE)

- Use LCD monitor as target. Display contrast pattern with varying grey levels to represent different contrast ratios, representing ratios from approximately 150:1.
- Set up target normal to Imaging Colorimeter. Distances: 0.5 m and 2 m. Measured from target to front of Imager, add 4.46 cm to account for distance to CCD plane ("Camera Distance").
- Use Electronic Lens control in Imaging Colorimeter software to adjust focus until best focus is achieved.
- Record "Focus Distance" value of the lens.
- Compare "Focus Distance" of lens to measured distance.
- Repeat for 10 different targets with 0.5 m and 2 m working distances.

RESULTS – PERCENT DEVIATION



Contrast Ratio (Luminance)	Grey values (Left/Right)	Measured Distance	Software Distance	Abs. Error (m)	% difference
139.40/255		0.548	0.542	-0.0056	1.02%
19.850/255		0.548	0.549	0.0014	0.26%
10.275/255		0.548	0.548	0.0004	0.07%
7.5100/255		0.548	0.542	-0.0056	1.02%
4.6128/255		0.548	0.543	-0.0046	0.84%
2.6165/255		0.548	0.543	-0.0046	0.84%
2.1185/255		0.548	0.543	-0.0046	0.84%
1.6205/255		0.548	0.543	-0.0046	0.84%
1.3225/255		0.548	0.542	-0.0056	1.02%
1.1245/255		0.548	0.542	-0.0056	1.02%

Contrast Ratio (Luminance)	Grey values (Left/Right)	Measured Distance	Software Distance	Abs. Error (m)	% difference
145.50/255		2.0486	2.045	-0.0036	0.2%
20.450/255		2.0486	2.046	-0.0026	0.1%
10.475/255		2.0486	2.048	-0.0006	0.0%
6.4100/255		2.0486	2.045	-0.0036	0.2%
4.1128/255		2.0486	2.046	-0.0026	0.1%
2.4165/255		2.0486	2.04	-0.0086	0.4%
2.0185/255		2.0486	2.055	0.0064	0.3%
1.6205/255		2.0486	2.041	-0.0076	0.4%
1.3225/255		2.0486	2.09	0.0414	2.0%
1.1245/255		2.0486	2.085	0.0364	1.8%

Figure B2

B.3 CONSIDERATIONS

B.3.1 Astigmatism

The measurement request is for the virtual image to be in focus. We use depth of field and technician decision.

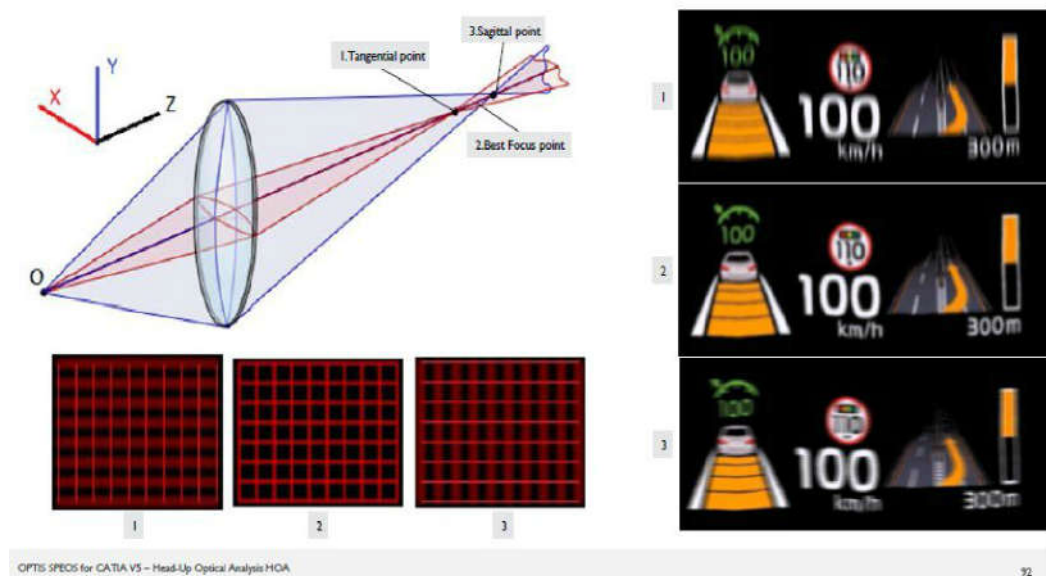
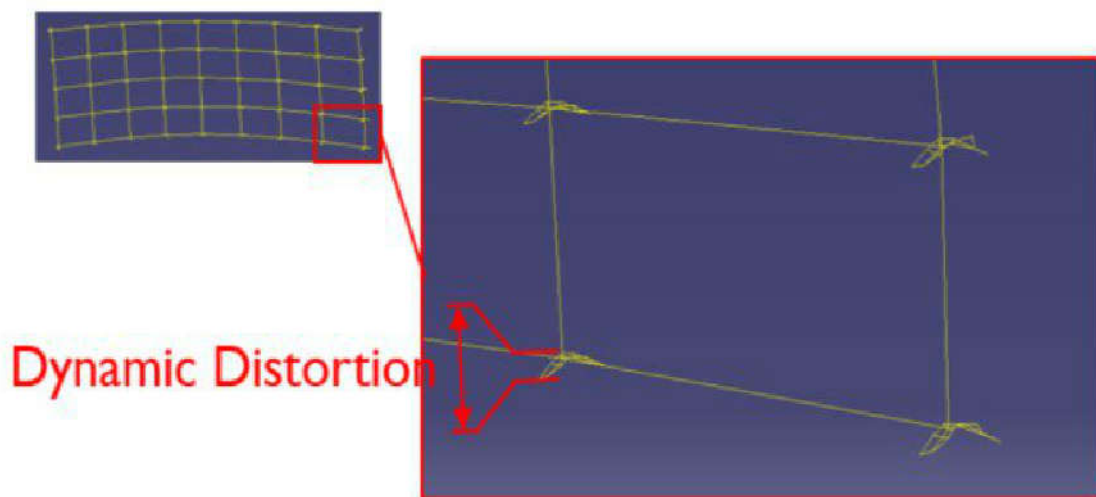


Figure B3

APPENDIX C

C.1 DISTORTIONS/ABERATIONS/EXAMPLES

*Figure C1*

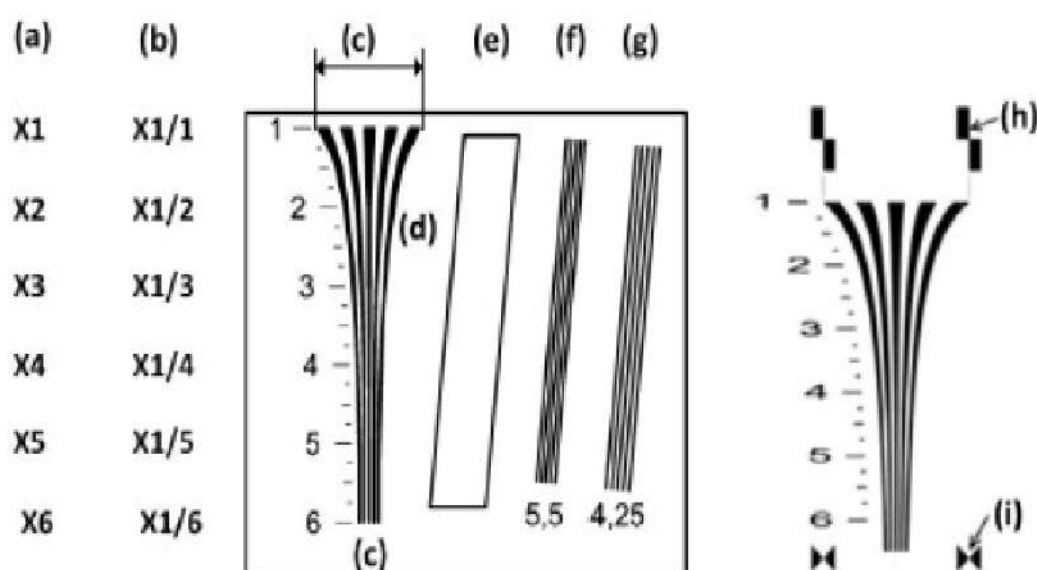
APPENDIX D

D.1 RESOLUTION MEASUREMENT

Commercial Display: Dot (pixels)/inch

ISO 12233: MTF = Modulation Transfer Function

$$\begin{aligned}
 \text{MTF}_{\text{sys}}(\xi, \eta) = & \text{MTF}_{\text{atmosphere}}(\xi, \eta) \cdot \text{MTF}_{\text{lens}}(\xi, \eta) \cdot \\
 & \text{MTF}_{\text{sensor}}(\xi, \eta) \cdot \text{MTF}_{\text{transmission}}(\xi, \eta) \cdot \\
 & \text{MTF}_{\text{display}}(\xi, \eta)
 \end{aligned}$$



Key

- (a) Multiplication factor of spatial frequency at respective position on the chart
- (b) Multiplication factor of line width at respective position on the chart
- (c) Width of 9 lines at reference position 1 on the chart
- (d) Area used to detect reference white level
- (e) Area used to detect reference black level or for SFR measurement
- (f) Exemplar line and space parallel bar chart with spatial frequency 5.5 times higher than reference position 1
- (g) Exemplar line and space parallel bar chart with spatial frequency 4.25 times higher than reference position 1
- (h) Example of reference width indicator (box cross edge symbol, 9-line width in this example)
- (i) Example of reference width indicator (triangular cross edge symbol, 9-line width in this example)

Figure D1 - Detail of a hyperbolic chart and examples of reference width indicator

NOTE: The hyperbolic charts which are available on the market indicate the nominal spatial frequency at reference position 1. The respective width of a single line at this reference position 1 equals to the image "HEIGHT" divided by the nominal spatial frequency given for position 1, independent of the chart orientation.

EXAMPLE: Chart with image HEIGHT = 1000 mm/PH and its hyperbolic chart at position 1 indicating 200 LW/PH means that a single line width at position (a) is:

$$1000 \text{ mm/PH} / 200 \text{ LW/PH} = 5 \text{ mm/LW}$$

At position 4, for example, the spatial frequency is increased four times and the width is reduced by 1/4, compared to the line at position 1, and the resulting line width at position 4 is 5/3 mm/LW.